pm which is shorter than the single bond distance of P-O (184 pm) due to pπ-dπ bonding and results in considerable double bond character.

In P_4O_{10} each P atoms form three bonds to oxygen atom and also an additional coordinate bond with an oxygen atom.

Terminal P-O bond length is 143 pm, which is less than the expected single bond distance. This may be due to lateral overlap of filled p orbitals of an oxygen atom with empty d orbital on phosphorous.

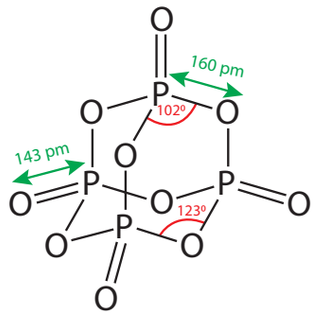
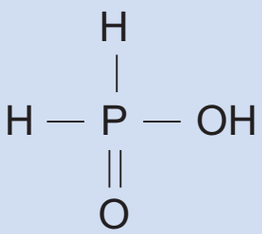
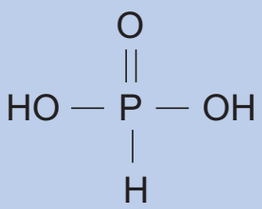
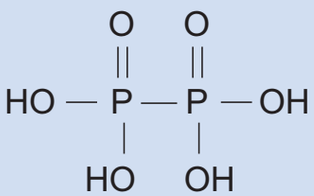
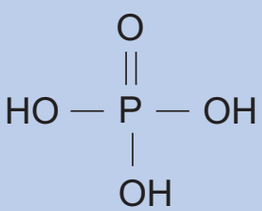
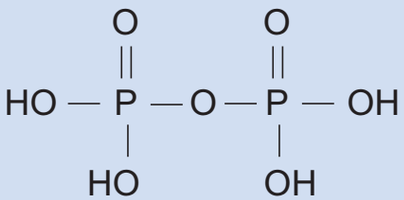


Figure 3.7 Structure of P_4O_{10}

Oxoacids of Phosphorous-Structure:

Name	Formula	Structure
Hypophosphorous acid	H_3PO_2	
Orthophosphrous acid	H_3PO_3	
Hypophosphoric acid	$H_4P_2O_6$	
Orthophosphoric acid	H_3PO_4	
Pyrophosphoric acid	$H_4P_2O_7$	

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